Lecture 9

$(X_1, X_2, \ldots, X_n)$ IID SAMPLE, $X_i \sim \text{Population}(\theta)$, WE ESTIMATE

θ WITH AN INTERVAL $[L_1, L_2]$ AT CONFIDENCE LEVEL $1-\alpha$

LOWER BOUND — UPPER BOUND

SUCH THAT $\mathbb{P}(L_1 \leq \theta \leq L_2) = 1-\alpha \quad \forall \theta \in \Theta$

EX: X: "PRODUCT WEIGHT". $X \sim N(\mu, \sigma^2)$

μ UNKNOWN

$\sigma = 2.5$ REASONABLE FROM LONG HISTORY OF OBS

MACHINE SHOULD BE CALIBRATED ON 250g.

IS THIS THE REALITY?

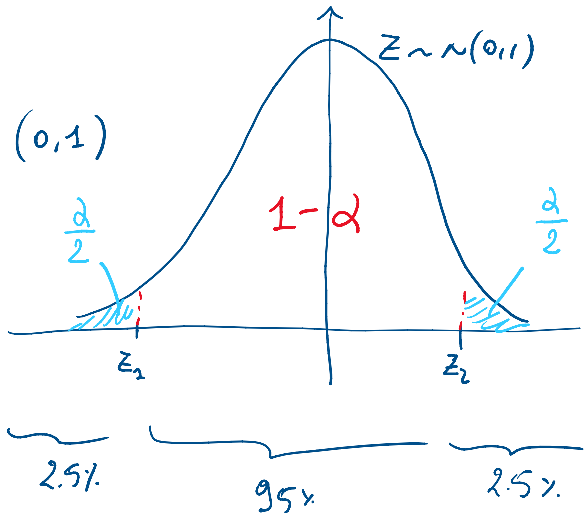
WE MAY DEVIATIONS FROM 250 g

- MACHINE CORRECTLY CALIBRATED, BUT SAMPLE VARIABILITY
- WRONG CALIBRATION

$\Rightarrow$ WE COLLECT $n = 25$ PRODUCTS $\Rightarrow \bar{X} = 250.2$ g SEEMS "REASONABLE"

" " " " $\Rightarrow \bar{X} = 251.1$ g " "

$\Rightarrow$ WE HAVE A SET OF REASONABLE VALUE THAT WILL FILL THE CI



$X \sim N(\mu, (2.5)^2) \Rightarrow \dfrac{X-\mu}{2.5} =: Z \sim N(0,1)$

$\mathbb{P}(z_1 < Z < z_2) = 1-\alpha$

FROM THIS, WE WANT TO REACH THE FORM $\mathbb{P}(L_1 \leq \theta \leq L_2) = 1-\alpha$

ALSO: WE KNOW HOW TO STANDARDIZE $\bar{X} \sim N\left(\mu, \frac{(2.5)^2}{n}\right)$

$Z := \dfrac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$

WE ARE GOING TOWARDS

$$Z := \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \sim N(0,1)$$

WE ARE GOING TOWARDS

$$\mathbb{P}(L_1 \leq \theta \leq L_2) = 1 - \alpha$$

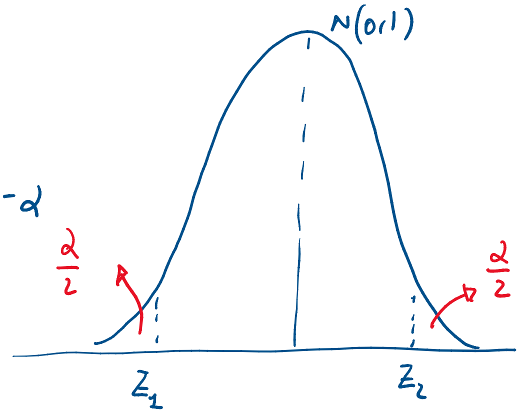
$$\mathbb{P}(z_1 \leq Z \leq z_2) = 1 - \alpha$$

$$\Rightarrow \mathbb{P}\left(z_1 \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_2\right) = 1 - \alpha$$

$$\Rightarrow \mathbb{P}\left(z_1 \frac{\sigma}{\sqrt{n}} \leq \bar{X} - \mu \leq z_2 \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

$$\Rightarrow \mathbb{P}\left(z_1 \frac{\sigma}{\sqrt{n}} - \bar{X} \leq -\mu \leq z_2 \frac{\sigma}{\sqrt{n}} - \bar{X}\right) = 1 - \alpha$$

$$\Rightarrow \mathbb{P}\left(\underbrace{\bar{X} - z_2 \frac{\sigma}{\sqrt{n}}}_{L_1} \leq \underbrace{\mu}_{\theta} \leq \underbrace{\bar{X} - z_1 \frac{\sigma}{\sqrt{n}}}_{L_2}\right) = 1 - \alpha$$



$\Rightarrow$ LOWER BOUND $L_1 = \bar{X} - z_2 \frac{\sigma}{\sqrt{n}}$

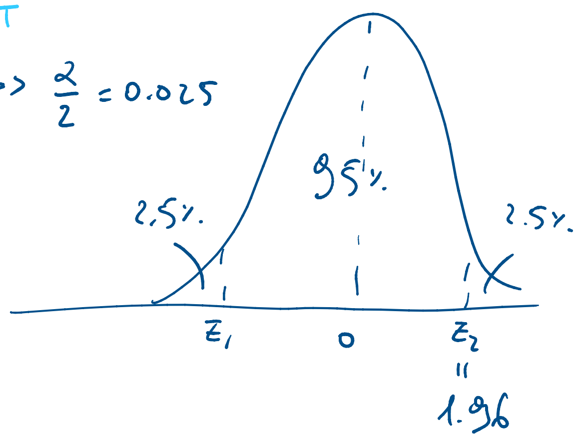
UPPER BOUND $L_2 = \bar{X} - z_1 \frac{\sigma}{\sqrt{n}}$

z_2 : QUANTILE OF $N(0,1)$ WITH $\frac{\alpha}{2}$ PROBABILITY ON ITS RIGHT,

DENOTED $\underbrace{z_{\frac{\alpha}{2}}}_{\text{AREA ON RIGHT}} = \text{qnorm}\big(\underbrace{1 - \tfrac{\alpha}{2}}_{\text{AREA ON THE LEFT}}\big)$

IF CONF. LEVEL $1 - \alpha = 0.95 \Rightarrow \alpha = 0.05 \Rightarrow \frac{\alpha}{2} = 0.025$

$$\Rightarrow z_2 = z_{0.025} = +1.96$$



z_1 : QUANTILE OF $N(0,1)$ WITH $\frac{\alpha}{2}$ ON ITS LEFT

$$\Rightarrow z_{1-\frac{\alpha}{2}} = \text{qnorm}\left(\frac{\alpha}{2}\right)$$

WITH $1 - \alpha = 0.95 \Rightarrow z_1 = z_{0.975} = -1.96$

$$L_1 = \bar{X} - z_2 \frac{\sigma}{\sqrt{n}} = \bar{X} - 1.96 \cdot \frac{2.5}{\sqrt{n}}$$

$$L_2 = \bar{X} - z_1 \frac{\sigma}{\sqrt{n}} = \bar{X} + 1.96 \cdot \frac{2.5}{\sqrt{n}}$$

$$\Rightarrow \boxed{CI_{95\%}(\mu) = \left[\bar{X} \mp 1.96 \frac{\sigma}{\sqrt{n}}\right] \quad \sigma \text{ KNOWN}}$$

$$L_2 = \bar{X} - z_1 \frac{\sigma}{\sqrt{n}} = \bar{X} + 1.96 \cdot \frac{2.5}{\sqrt{n}}$$

95% ... $\sqrt{n}$]

σ KNOWN

$X \sim N(\mu, \sigma^2)$

NOTE THAT CI HAS THE FORM $[\bar{X} \mp$ MARGIN OF ERROR$]$

POINT ESTIMATOR

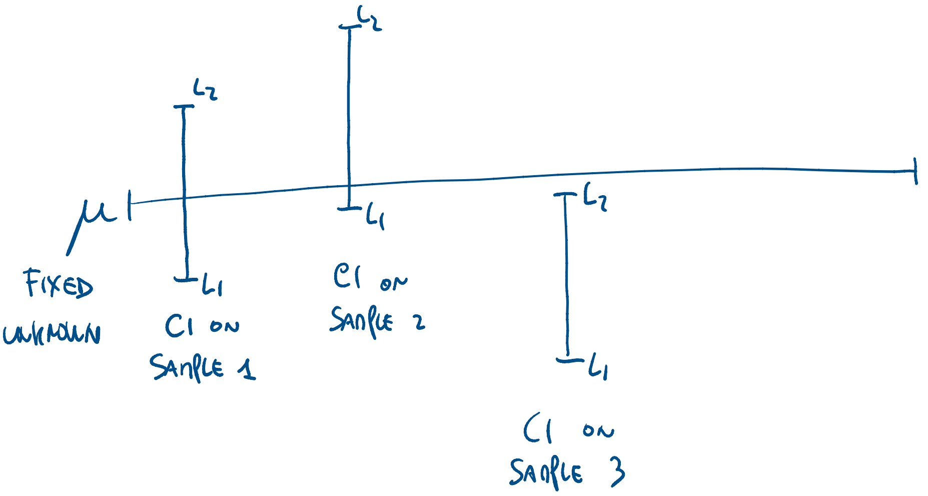
IN THE EXAMPLE, THE REALIZED VALUE OF $CI_{95\%}(\mu) = \left[250.2 \mp 1.96 \cdot \frac{2.5}{\sqrt{25}}\right]$

$= [249.22, 251.18]$

$\Rightarrow$ THE ASSUMED 250g VALUE IS PART OF THE CI $\Rightarrow$ 250g IS REASONABLE FOR μ

(AMONG REASONABLE VALUES FOR μ)

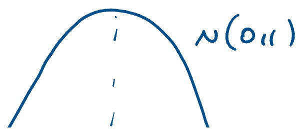
$\Rightarrow$ WE SAY THAT THE OBSERVED DEVIATION FROM 250g IS DUE TO SAMPLE VARIABILITY AND NOT TO WRONG CALIBRATION.



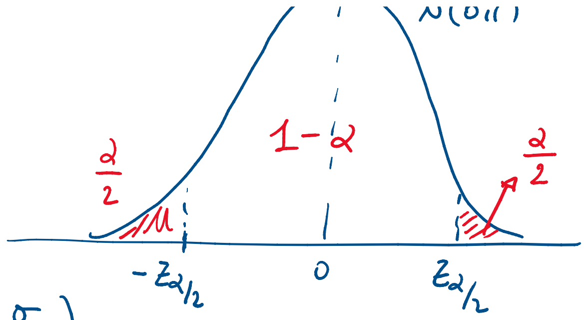
$(1-\alpha)$ PROPORTION OF CIs WILL INCLUDE THE TRUE μ

IF WE GENERALIZED PREVIOUS CI TO GENERAL $1-\alpha$, WE HAVE

$$\boxed{CI_{1-\alpha}(\mu) = \left[\bar{X} \mp z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right]}$$

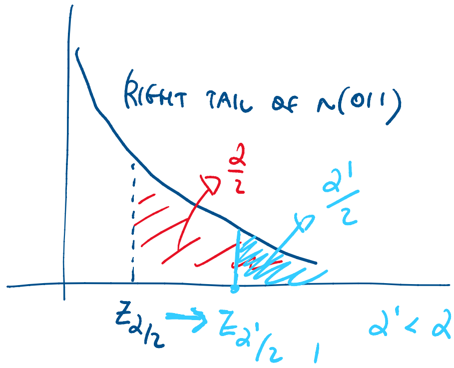


$$\boxed{\begin{array}{c} CI_{1-\alpha}(\mu) = \left[\bar{X} \mp z_{\alpha/2} \dfrac{\sigma}{\sqrt{n}}\right] \\ X \sim N(\mu, \sigma^2),\ \sigma^2 \text{ KNOWN} \end{array}}$$



WIDTH : $A = L_2 - L_1 = \left(\bar{X} + z_{\alpha/2}\dfrac{\sigma}{\sqrt{n}}\right) - \left(\bar{X} - z_{\alpha/2}\dfrac{\sigma}{\sqrt{n}}\right)$

$$= 2\, z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$



- IF CONF LEVEL $\uparrow \Rightarrow (1-\alpha)\uparrow \Rightarrow \alpha \downarrow \Rightarrow z_{\alpha/2}\uparrow$

  $\Rightarrow A = 2 z_{\alpha/2} \dfrac{\sigma}{\sqrt{n}} \uparrow$ WIDER CI (FREE TO MODIFY)

- IF $n \uparrow \Rightarrow A \downarrow$ STRICTER CI (SOMETIMES POSSIBLE)

- IF $\sigma^2 \uparrow \Rightarrow A \uparrow$ WIDER CI (NOT UNDER CONTROL)

WE CONSTRUCT CIs WITH THE METHOD OF PIVOTAL QUANTITY (PQ)

A PQ IS

- A FUNCTION OF $\underline{X}$ AND OF PAR FOR WHICH I WANT THE CI
  (IN THE EX: OF $\underline{X}$ AND OF μ)
- ITS DISTRIBUTION DOES NOT DEPEND ON THE PARAMETER OF INTEREST
  (IN THE EX: $Z \sim N(0,1)$ DOES NOT DEPEND ON μ)
- INVERTIBLE WRT THE PAR OF INTEREST
  (IN THE EX: I CAN SOLVE FOR μ).

USUALLY: THE PQ FOR A LOCATION PARAMETER HAS THE FORM

$$PQ = \frac{\hat{\theta} - \theta}{\hat{\sigma}_{\hat{\theta}}} = \frac{\text{POINT ESTIMATOR} - \text{PAR}}{\text{ESTIMATED STD OF ESTIMATOR}}$$

(IN THE EX: PQ WAS $Z = \dfrac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$)

(IN THE EX: PQ WAS $Z: \underbrace{\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}}_{\substack{1)\ \text{DEPENDS ON } \bar{X} \text{ AND ON } \mu \\ 3)+\ \text{INVERTIBLE FOR } \mu}} \sim \underbrace{N(0,1)}_{2)\ \text{DOES NOT DEPEND ON } \mu}$)

REMARK: IF WE HAVE $[L_1, L_2]$ CI OF θ AT $1-\alpha$ LEVEL, THEN I ALSO HAVE A CI FOR $g(\theta)$, WHERE g IS MONOTONE FUNCTION.

EX: $CI_{1-\alpha}(\sigma^2) = [L_1, L_2]$

$\Rightarrow CI_{1-\alpha}(\sigma) = [\sqrt{L_1}, \sqrt{L_2}]$, SINCE $\sigma = g(\sigma^2) = \sqrt{\sigma^2}$ CI OF SD

$\Rightarrow CI_{1-\alpha}(1/\sigma^2) = \left[\frac{1}{L_2}, \frac{1}{L_1}\right]$, SINCE $g(\sigma^2) = 1/\sigma^2$ CI OF PRECISION

EX:

$X \sim N(\mu, \sigma^2)$, $\sigma^2 = 400$. WHAT IS THE MIN n TO COLLECT TO HAVE A $CI_{95\%}(\mu)$ OF WIDTH BELOW 10?

$1-\alpha = 0.95 \Rightarrow \alpha = 0.05$

$$A = L_2 - L_1 = 2\, z_{\alpha/2} \frac{\sigma}{\sqrt{n}} = 2 \cdot \underbrace{z_{0.025}}_{\substack{\text{qnorm}(0.975) \\ = 1.96}} \cdot \frac{\sqrt{400}}{\sqrt{n}} \leq 10$$

THEN $2 \cdot 1.96 \cdot \frac{\sqrt{400}}{\sqrt{n}} \leq 10 \Rightarrow n \geq \left(\frac{2 \cdot 1.96 \cdot \sqrt{400}}{10}\right)^2 = 61.463$

$\Rightarrow$ WE NEED TO COLLECT AT LEAST 62 OBS.

NOW: WE COLLECT $n = 64$ OBS, $\sum_{i=1}^{n} X_i = 468$. FIND $CI_{95\%}(\mu)$.

$$\left[\bar{X} \mp z_{0.025} \frac{\sigma}{\sqrt{n}}\right] = \left[\frac{468}{64} \mp 1.96 \cdot \frac{\sqrt{400}}{\sqrt{64}}\right] = [4.1, 16.9]$$

$$\left[\bar{X} \mp z_{0.025} \frac{\sigma}{\sqrt{n}}\right] = \left[\frac{768}{64} \mp 1.96 \cdot \frac{\sqrt{400}}{\sqrt{64}}\right] = [7.1, 16.9]$$

$A = L_2 - L_1 = 16.9 - 7.1 = 9.8 < 10$ AS EXPECTED SINCE $n > 62$

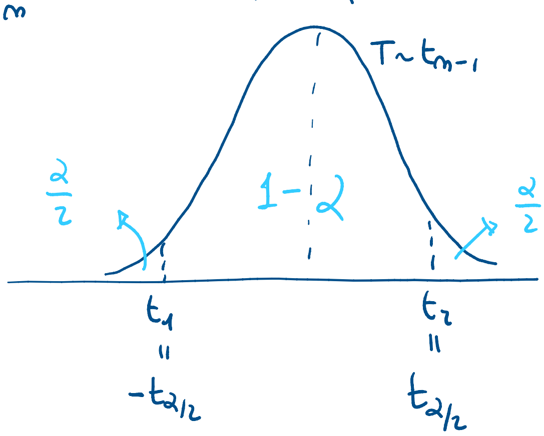
$\boxed{CI_{1-\alpha}(\mu),\ X \sim N(\mu, \sigma^2),\ \sigma^2 \text{ UNKNOWN}}$

$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$ CAN'T BE USED, SINCE I CAN'T COMPUTE THE DENOMINATOR (I DON'T KNOW σ)

$\Rightarrow$ REPLACE σ WITH S

$\Rightarrow$ PQ IS NOW $T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$ (T-STUDENT WITH $n-1$ DF)

$$P(t_1 < T < t_2) = 1 - \alpha$$



QUANTILE OF t_{n-1} WITH $\frac{\alpha}{2}$ ON RIGHT

$$\Rightarrow P(-t_{\alpha/2} < T < +t_{\alpha/2}) = 1 - \alpha$$

$$\Rightarrow P\left(-t_{\alpha/2} < \frac{\bar{X} - \mu}{S/\sqrt{n}} < +t_{\alpha/2}\right) = 1 - \alpha$$

$$\Rightarrow P\left(-t_{\alpha/2} \frac{S}{\sqrt{n}} \leq \bar{X} - \mu \leq t_{\alpha/2} \frac{S}{\sqrt{n}}\right) = 1 - \alpha$$

$$\Rightarrow P\Big(\underbrace{\bar{X} - t_{\alpha/2} \frac{S}{\sqrt{n}}}_{L_1} \leq \underbrace{\mu}_{\theta} \leq \underbrace{\bar{X} + t_{\alpha/2} \frac{S}{\sqrt{n}}}_{L_2}\Big) = 1 - \alpha$$

$$\Rightarrow \boxed{CI_{1-\alpha}(\mu) = \left[\bar{X} \mp t_{\alpha/2} \frac{S}{\sqrt{n}}\right]}$$

WHERE $t_{\alpha/2} = qt\left(1 - \frac{\alpha}{2}, n-1\right)$

$$A = L_2 - L_1 = 2 \cdot t_{\alpha/2} \frac{S}{\sqrt{n}}$$

- AS BEFORE: IF $n \uparrow$, $\underbrace{A \downarrow}_{\text{ON AVERAGE CORRECT}}$, BUT NOW A IS A RV

- σ UNKNOWN $\Rightarrow$ INCREASED UNCERTAINTY $\Rightarrow$ AT A GIVEN $1-\alpha$, $t_{\alpha/2}$ IS HIGHER THAN $z_{\alpha/2}$